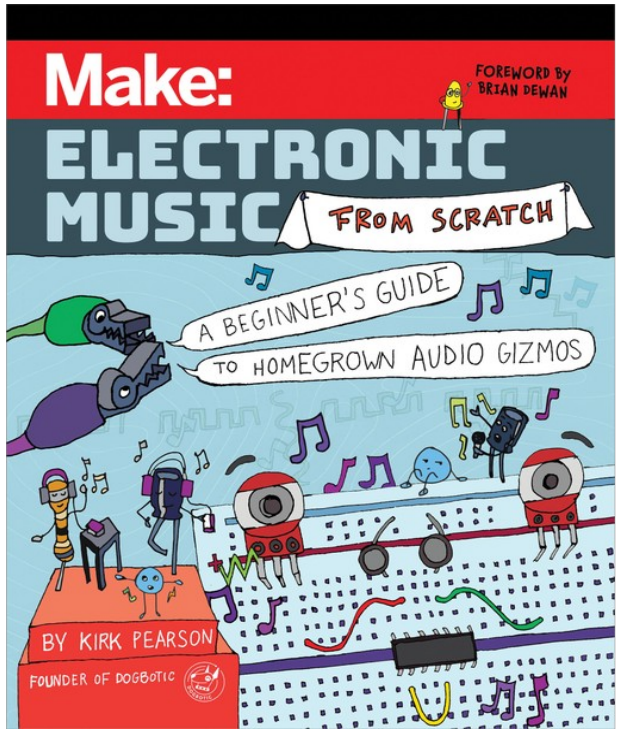
Electronic Music from Scratch

Reference: https://dogbotic.com/merch/book

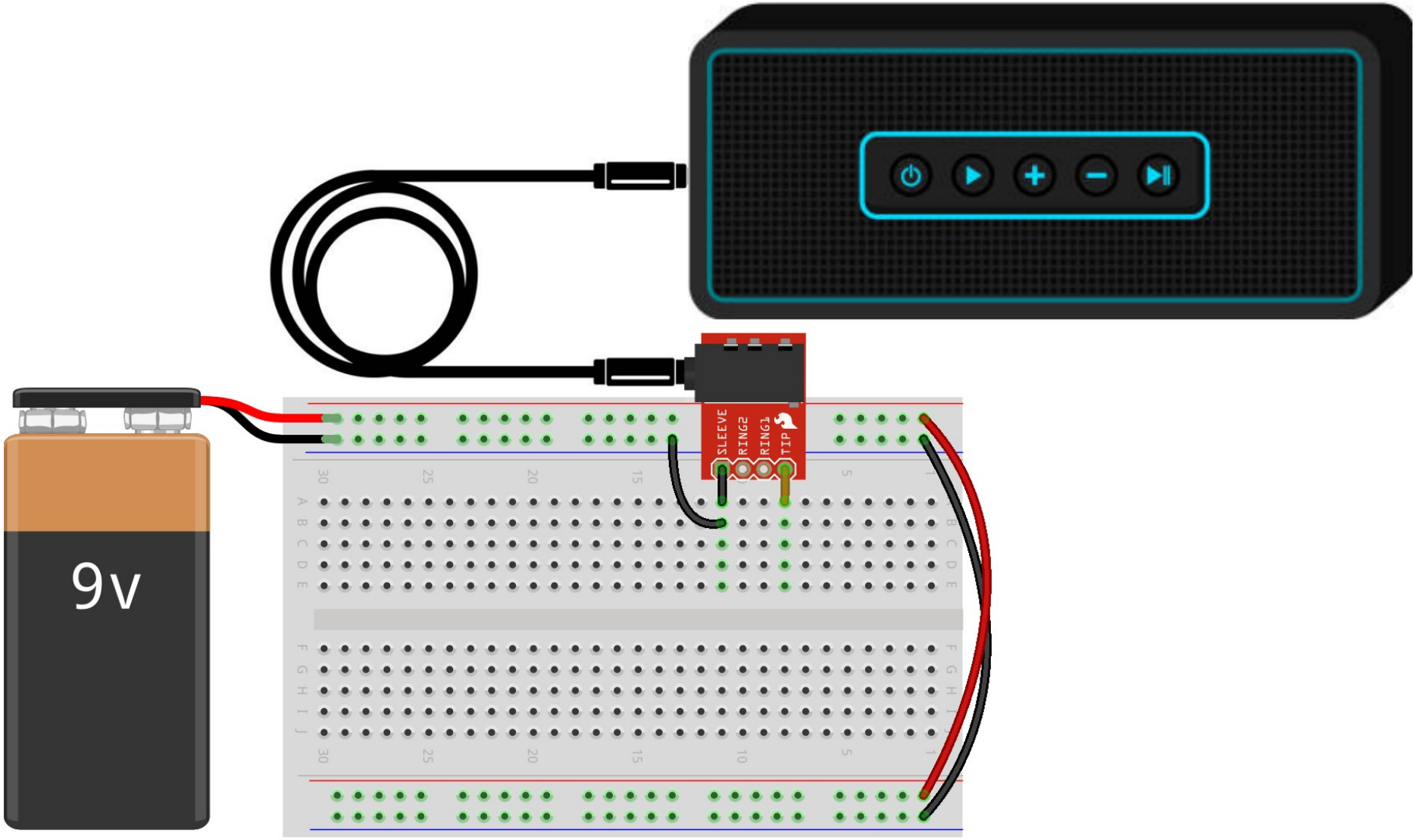
“A crash course in the joys of musical circuitry that teaches makers to become musicians, and musicians to become makers. Written for total beginners, this approachable guide uses more than 40 hands-on experiments and projects to build whimsical, wild, and just plain weird gizmos.”



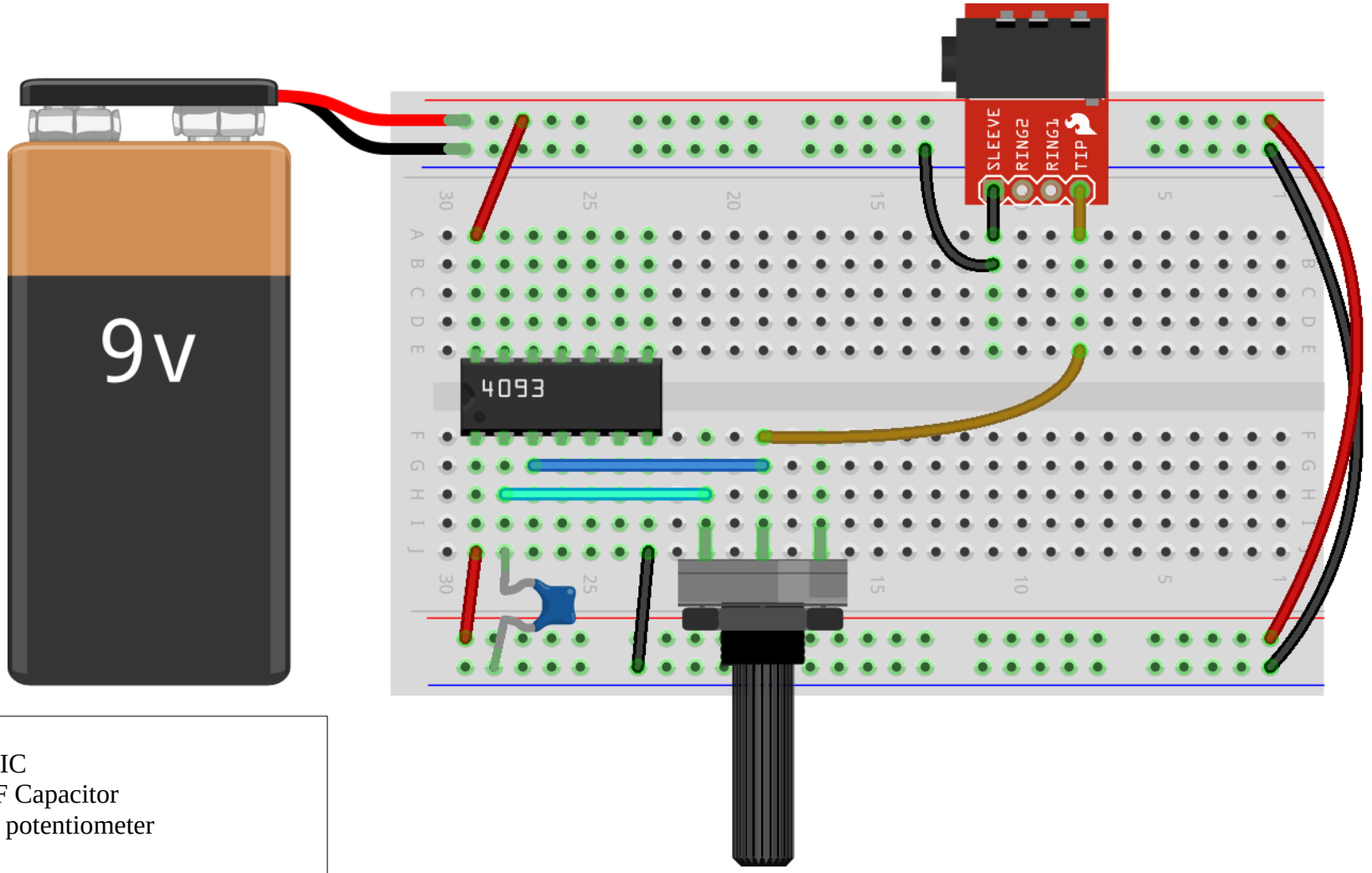
Basic Kit List

- Solderless Breadboard.
- Jumper wires.
- Range of resistors and capacitors.
- Range of integrated circuits (“chips”).
- Potentiometers.
- TRS 3.5mm “jack” socket breakout board.
- 9V battery and clip.
- Headphone lead, amplification and speaker.

Basic Setup

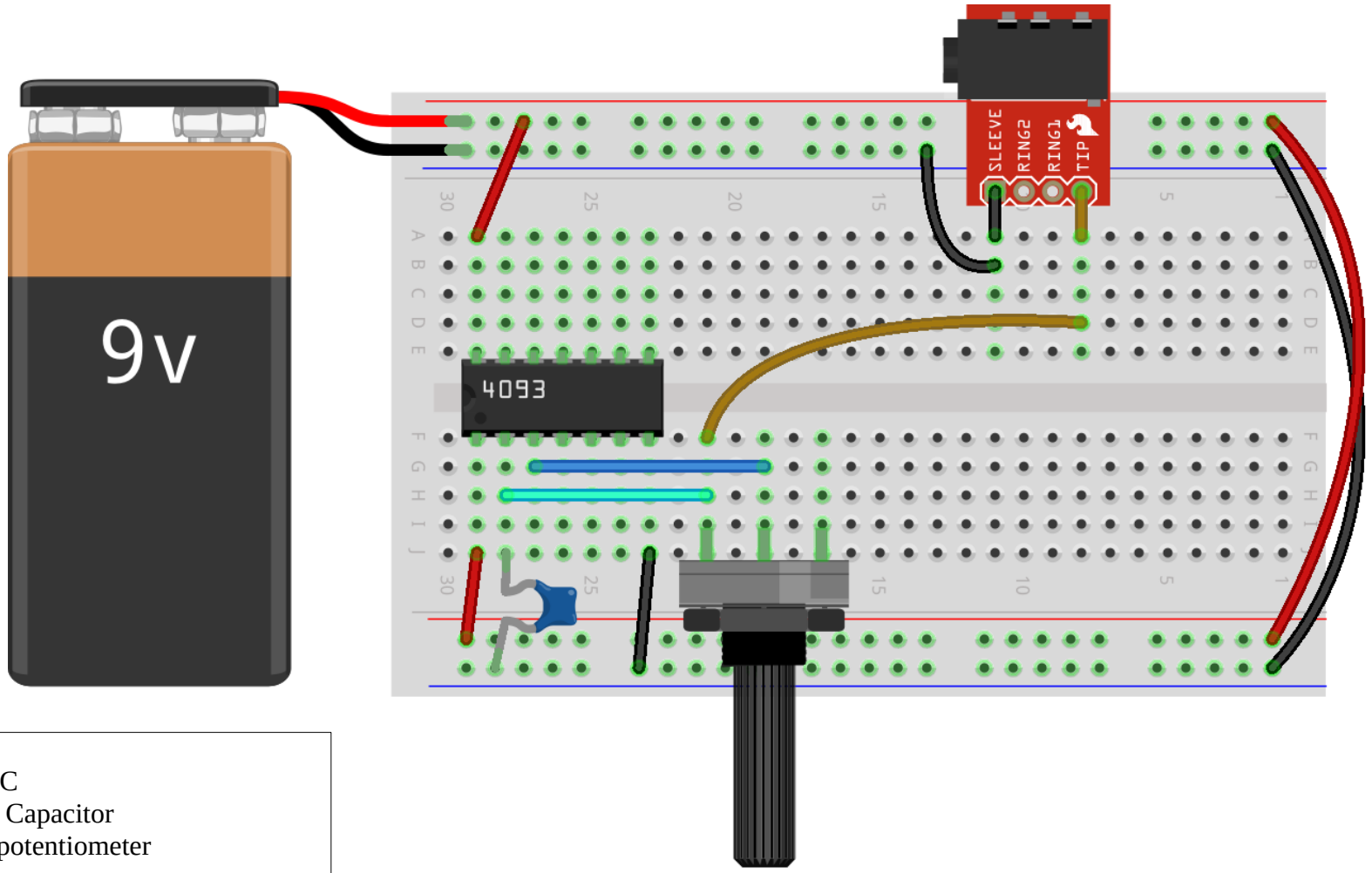


Square Wave Oscillator



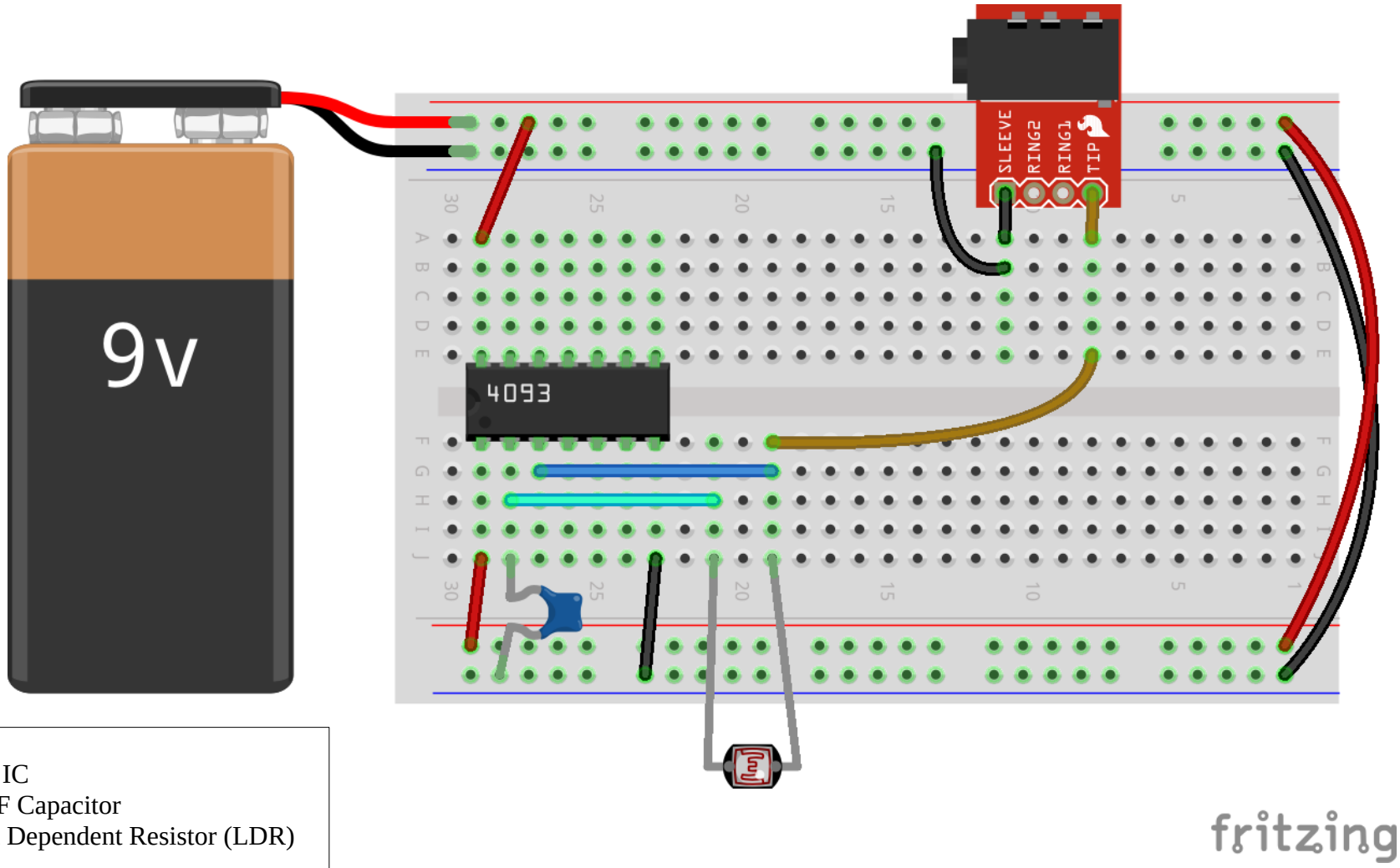
Parts:
- 4093 IC
- 100nF Capacitor
- 100K potentiometer

Triangle Wave Oscillator



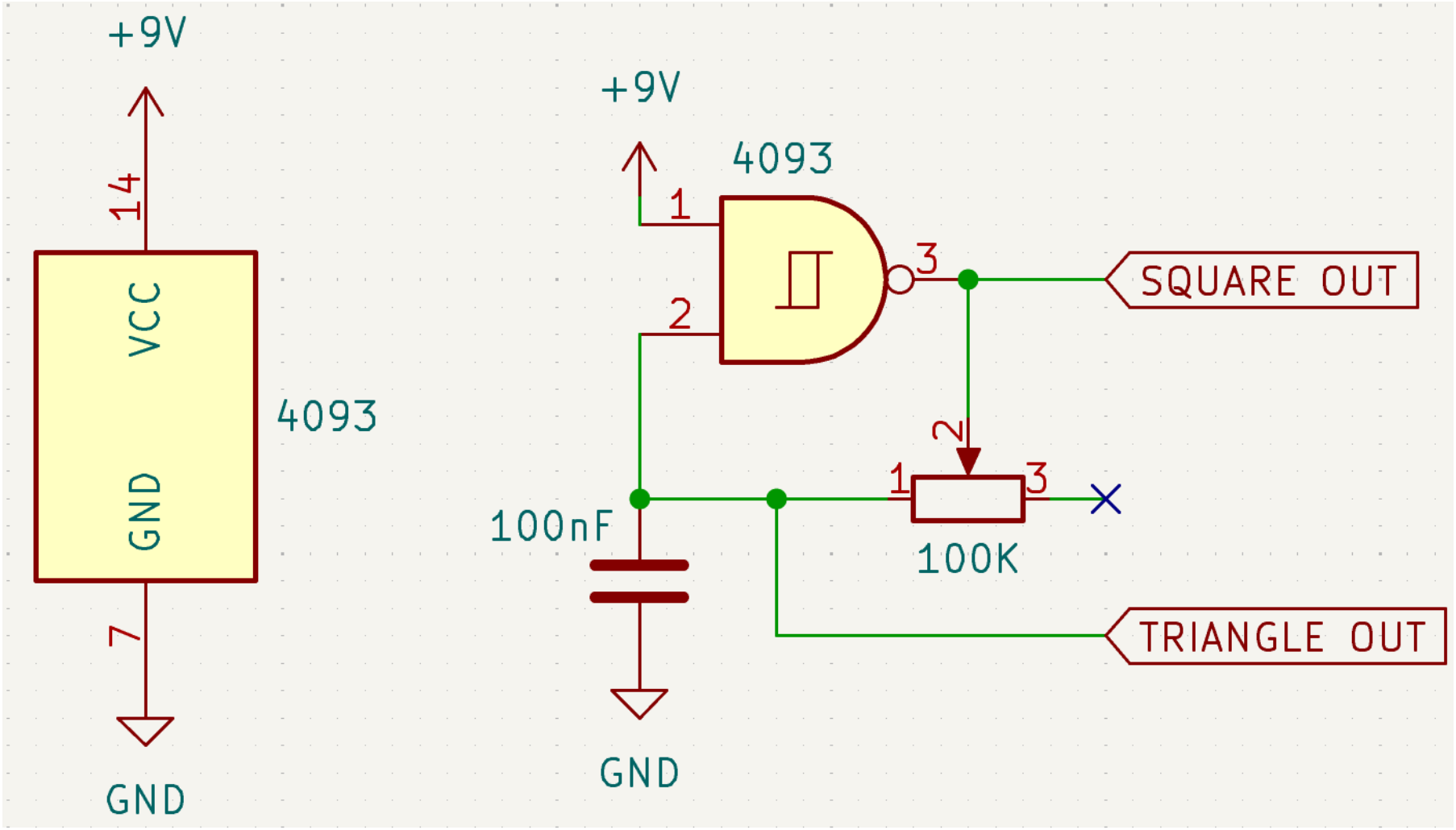
Parts:
- 4093 IC
- 100nF Capacitor
- 100K potentiometer

Light Controlled Oscillator

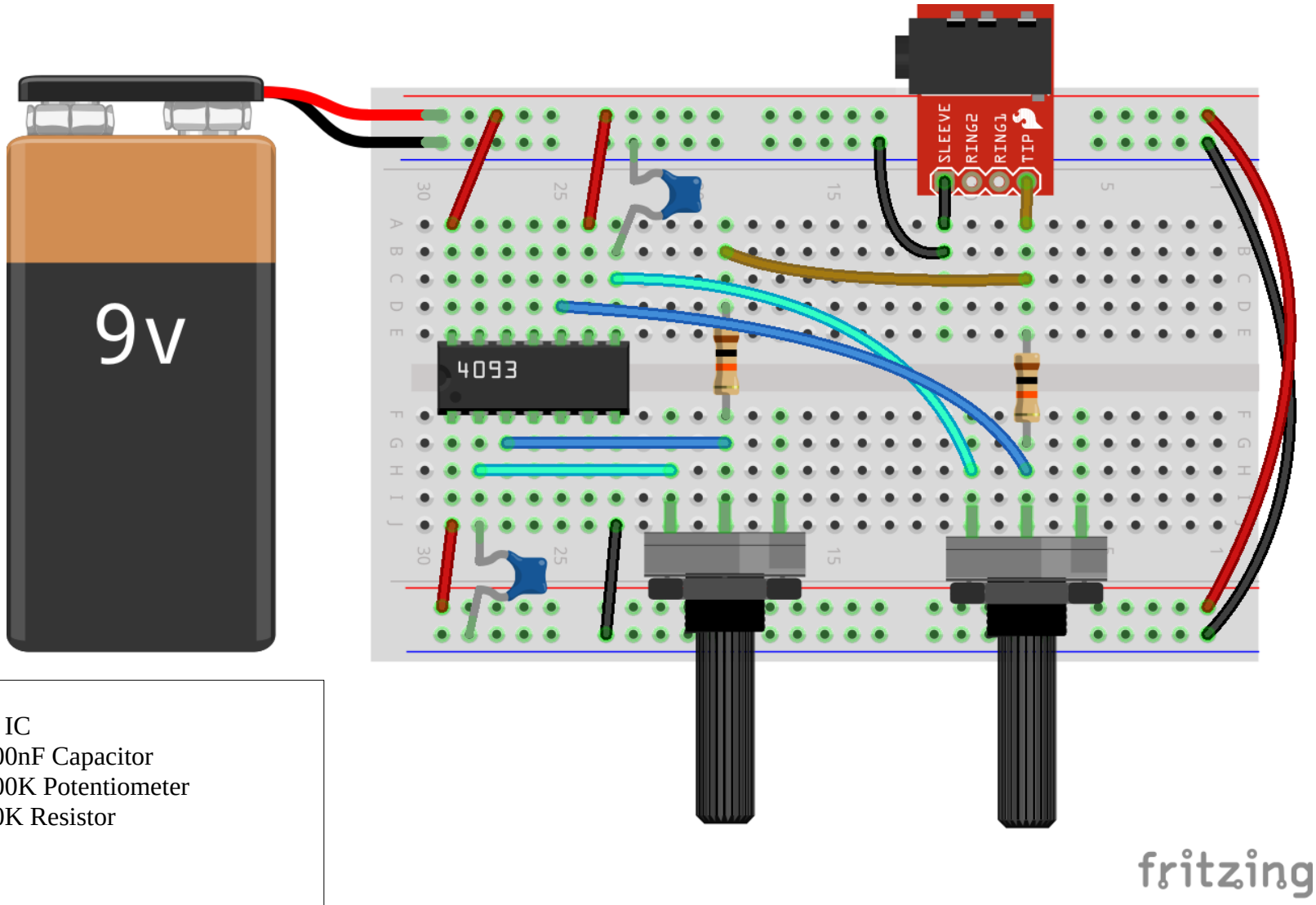


Parts:
- 4093 IC
- 100nF Capacitor
- Light Dependent Resistor (LDR)

Oscillator Circuit Diagram

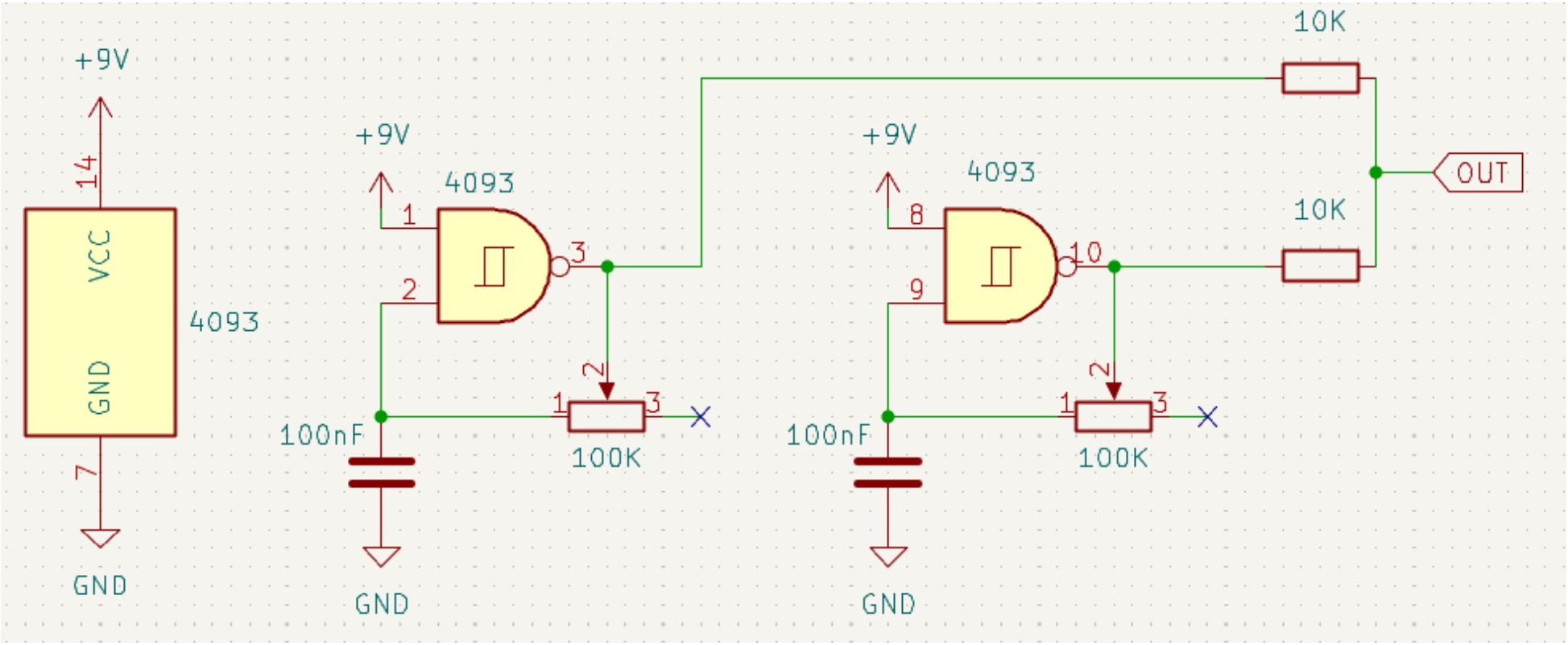


Poly Oscillator

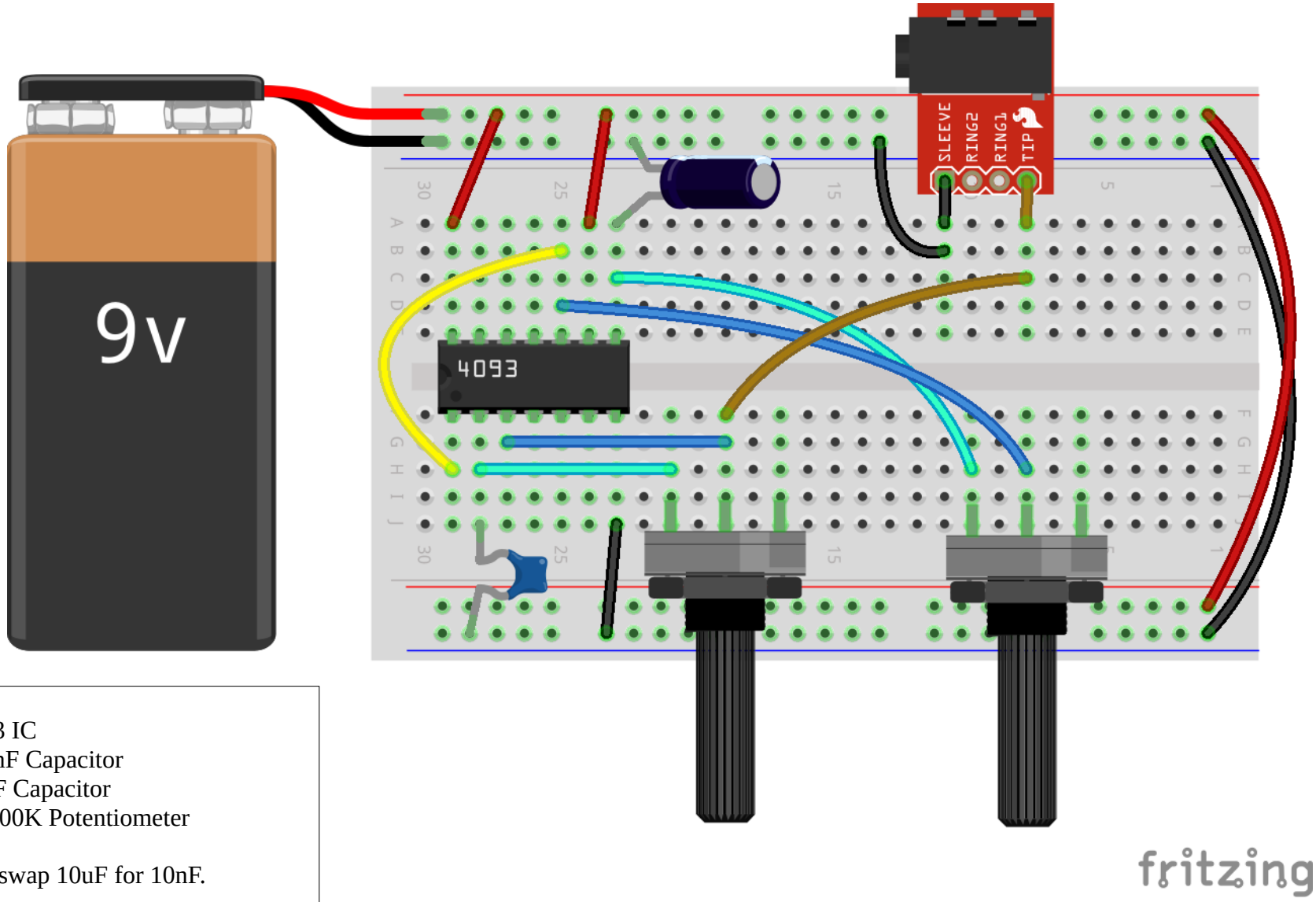


Parts:
- 4093 IC
- 2x 100nF Capacitor
- 2x 100K Potentiometer
- 2x 10K Resistor

Poly Oscillator Circuit Diagram



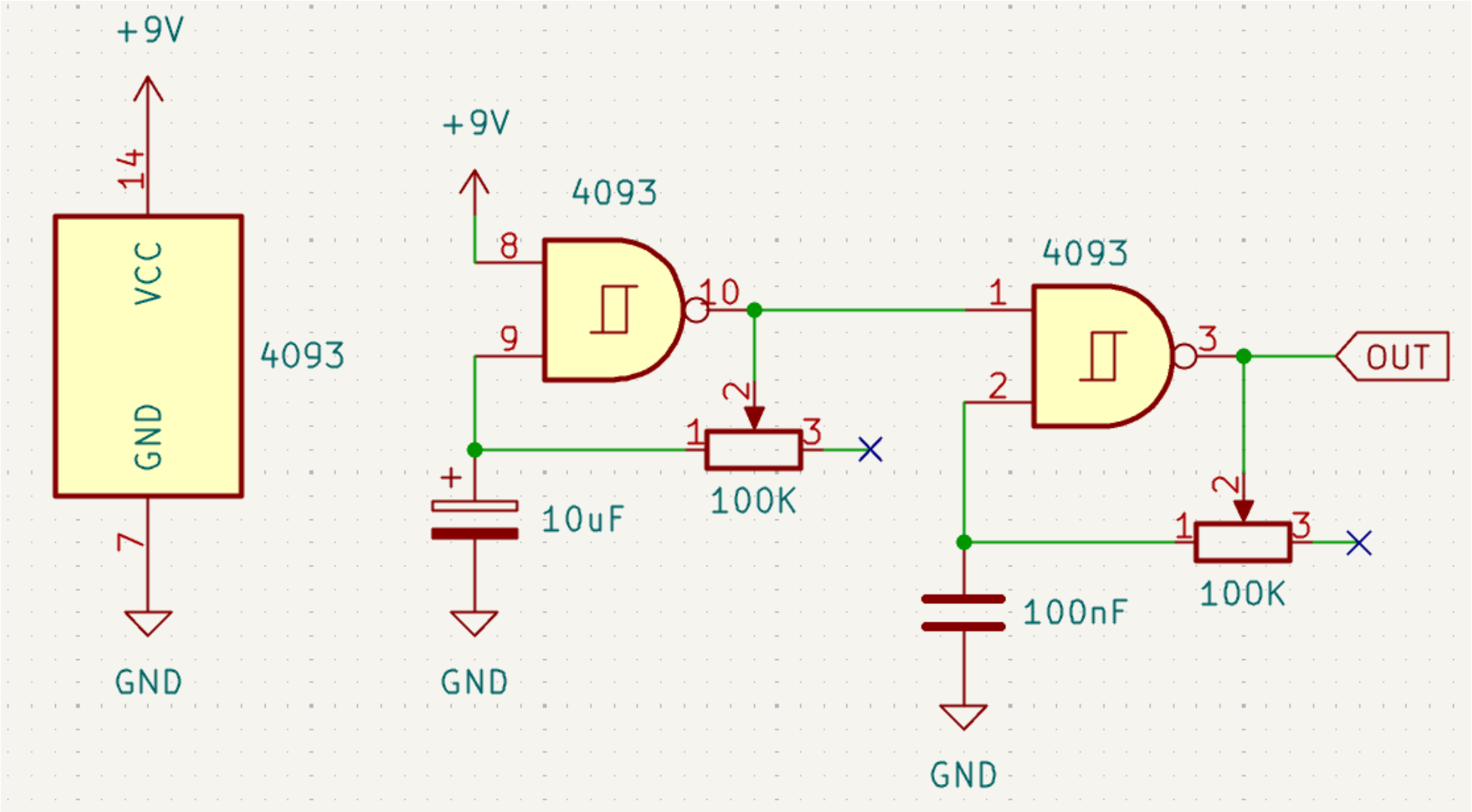
Tremolo Oscillator



Parts:
- 4093 IC
- 100nF Capacitor
- 10uF Capacitor
- 2x 100K Potentiometer

Then swap 10uF for 10nF.

Tremolo Oscillator Circuit Diagram



Next Experiment